



Amazon Prime Video Cloud Computing Case Study

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About Amazon Prime Video

Amazon Prime Video is a global on-demand streaming platform delivering movies, TV shows, sports, and original content to millions of concurrent users worldwide. Cloud computing is not optional — it is the operational foundation enabling this scale.

Massive Scale

Millions of concurrent viewers generating unpredictable traffic spikes

Global Reach

Geographically distributed audiences requiring low-latency delivery

High Bandwidth

Large video libraries with multiple resolutions and formats

Always On

High availability and fault tolerance required 24/7



Cloud Model Used by Prime Video



Deployment Model

Prime Video operates on **AWS Public Cloud** — a globally distributed infrastructure providing elastic, on-demand resources without physical data centre ownership.

Service Models

- **IaaS** — EC2 compute, VPC networking, EBS storage
- **PaaS** — Managed services (RDS, Lambda, ECS) without infrastructure overhead
- **SaaS** — Prime Video itself, delivered as a cloud application to end users

Why Prime Video Uses Cloud Computing

Each operational challenge maps directly to a cloud capability.

Challenge	Cloud Solution
Millions of concurrent users	Elastic scalability
Traffic spikes	Auto Scaling
Global audience	Distributed infrastructure
Large video library	Cloud object storage
High availability	Redundancy across AZs
Fast streaming	CDN / Edge delivery
Security & compliance	IAM, encryption, monitoring
Disaster recovery	Geo-distributed backups

 **Key Insight:** Cloud enables Prime Video to scale dynamically while maintaining availability and streaming performance.

Prime Video Cloud Architecture

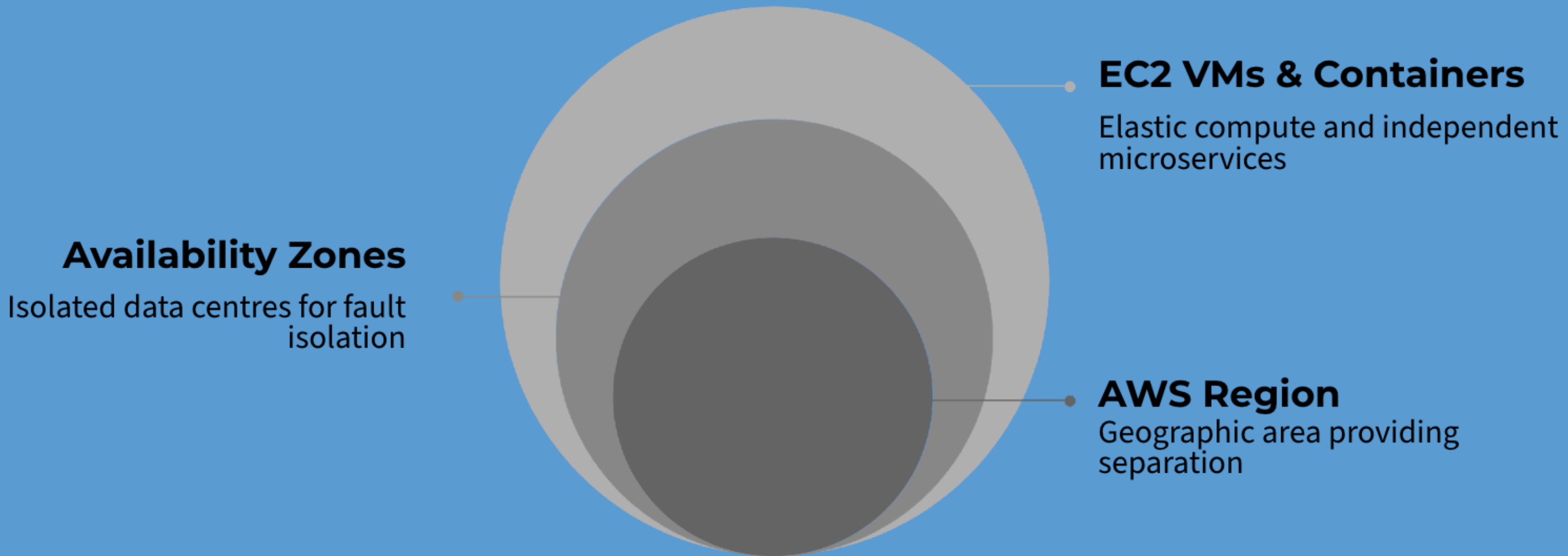


Key AWS Components

- **EC2** — Elastic compute for application workloads
- **Elastic Load Balancing** — Distributes traffic across instances
- **Auto Scaling** — Adjusts capacity to real-time demand
- **AWS Regions & AZs** — Geographic redundancy
- **Amazon CloudFront** — Global CDN for edge delivery

Where specific Prime Video service usage is not publicly confirmed, AWS capabilities are presented as architecturally relevant examples.

Data Centres, VMs, Containers & Networking



Regions provide geographic separation; Availability Zones provide fault isolation. EC2 VMs deliver elastic compute, while containers and microservices enable independent scaling of individual application components.

Load Balancing

Distributes viewer requests across healthy instances to prevent overload

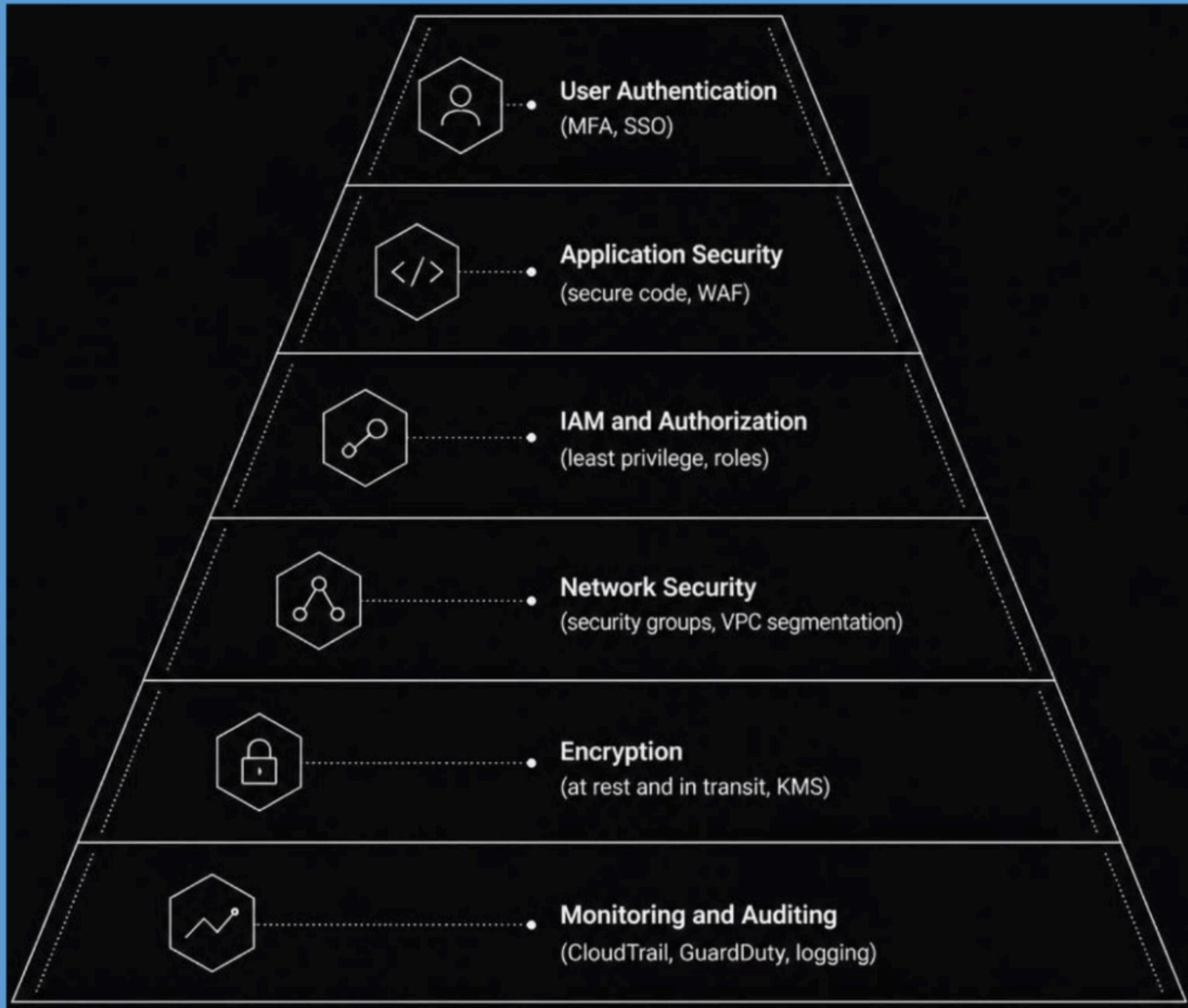
CDN & Edge Caching

CloudFront caches video segments at edge locations, reducing latency for geographically dispersed viewers

Network Redundancy

Multi-AZ deployment ensures continuity during infrastructure failures

Security Architecture



Defence in Depth

Prime Video applies layered security across identity, application, network, and data layers.

- **IAM** — Fine-grained access control and least-privilege policies
- **Encryption** — TLS in transit; AES-256 at rest via AWS KMS
- **Network Security** — Security groups, VPC isolation, WAF
- **Monitoring** — CloudTrail logging, GuardDuty threat detection

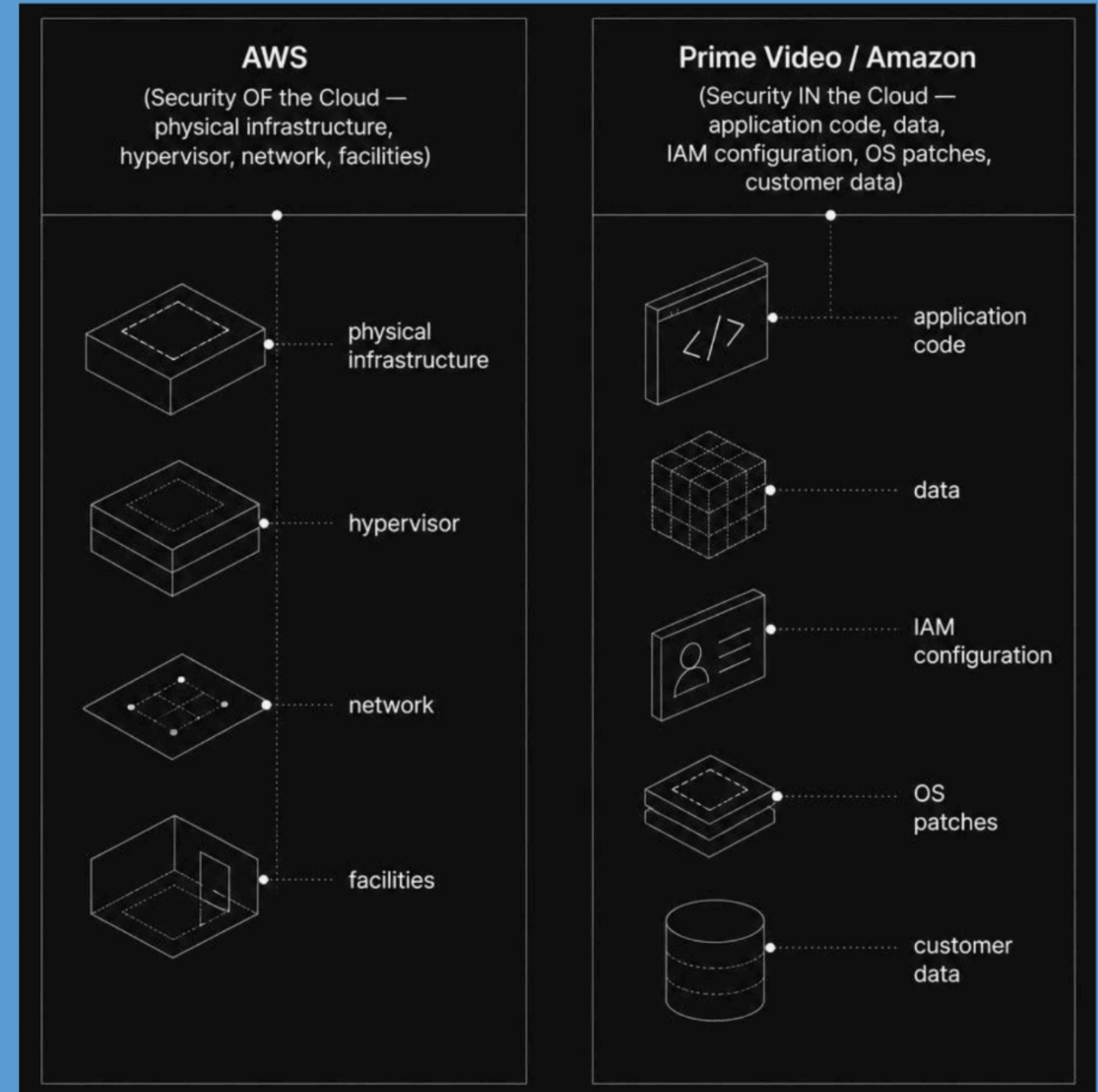
Privacy, Compliance & Shared Responsibility

Compliance Standards

- **ISO/IEC 27001** — Information security management
- **SOC 2** — Security and availability controls
- **PCI DSS** — Payment data (where applicable)
- **GDPR** — EU data protection requirements

AWS compliance certifications do not automatically apply to every Prime Video workload independently.

Shared Responsibility Model



Video Storage & Data Management

Content Creation

Encoding & Processing

Amazon S3 Storage

CloudFront CDN

More viewers and content trigger automatic scaling across distributed infrastructure.

Object Storage

Amazon S3 — Durable, scalable storage for video assets and encoded files

Block Storage

Amazon EBS — Persistent volumes for EC2 compute workloads

Databases

DynamoDB and **RDS** — User profiles, metadata, viewing history, recommendations

☐ Prime Video operates at massive data scale; exact database size is not publicly disclosed.

Real-World Cloud Benefits

Cloud Capability	Benefit to Prime Video
Elasticity	Handles sudden demand
Global Infrastructure	Supports worldwide users
CDN	Faster video delivery
Cloud Storage	Scales with content library
Distributed Architecture	High availability
Security Services	Protects users and content
Managed Services	Reduces infrastructure management
Monitoring	Detects failures and threats

Scalability	Availability	Performance
Security	Cost Efficiency	

What Prime Video Demonstrates About Cloud Computing

- 1 Cloud enables global streaming at scale
- 2 AWS provides compute, networking, and storage
- 3 Distributed architecture improves availability and performance
- 4 CDN and edge delivery reduce streaming latency
- 5 Security mechanisms protect users, applications, and content
- 6 Cloud storage and databases support massive data
- 7 Elasticity allows infrastructure to adapt to changing demand

"Prime Video is a real-world example of how cloud computing transforms a high-scale digital service into a globally available, scalable and resilient platform."